WASP-12b

R-0566

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_170123` · created 2026-07-23T21:18:11+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457776.68735 +/- 0.00034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.045
Transit depth (Rp/Rs)^2	2.36 +/- 1.37 [%]
Orbital Inclination (inc)	86.15 +/- 2.89
Airmass coefficient 1 (a1)	0.075 +/- 0.025
Airmass coefficient 2 (a2)	2.55 +/- 1.37
Scatter in the residuals of the lightcurve fit is	38.32 %
Best Comparison Star	#2 - [301, 400]
Optimal Aperture	61.5
Optimal Annulus	61.5
Transit Duration (day)	0.0012 +/- 0.0018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.045 vs 0.1178 (deviation 0.67σ)
Duration fit vs published	0.0012 d vs 0.125 d (deviation 59.09σ)
Mid-time O–C	-0.9 min
Depth SNR	2.9
Residual scatter	38.32 %

Light curve

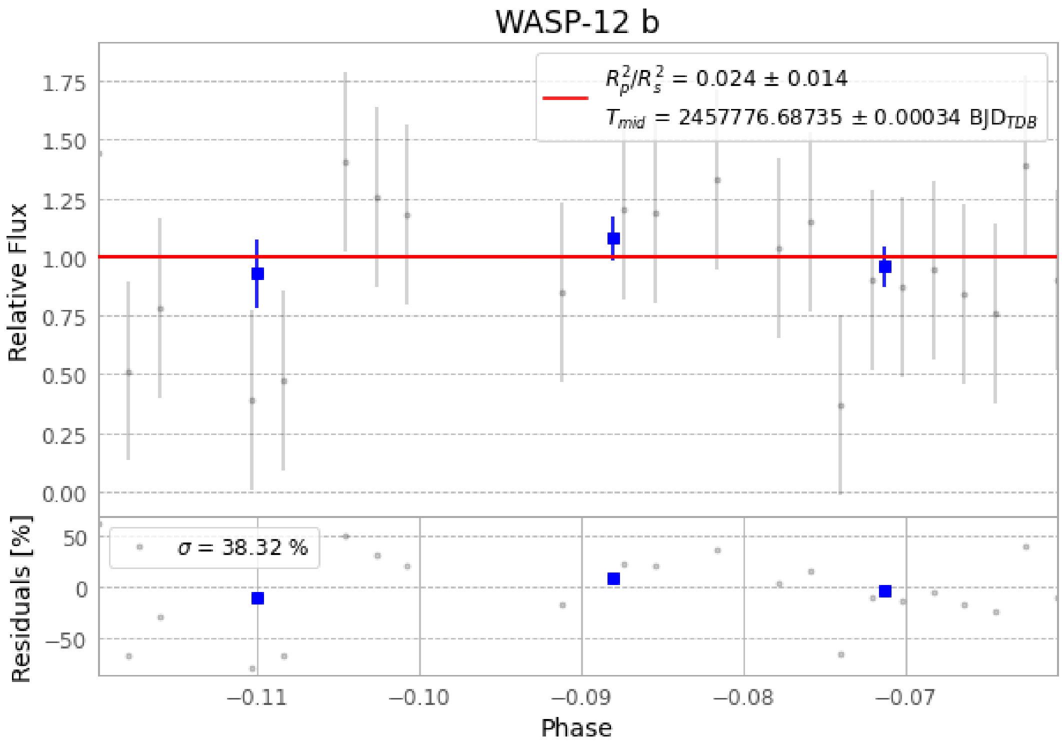

Figure 1 — FinalLightCurve_WASP-12 b_2017-01-23.png (SHA-256 c2ae05d60e261a6b…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [244, 428], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6d28aae8be18f967…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

32 FITS frames (21 MB), `WASP-12170124032412.FITS` → `WASP-12170124045712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-170123.log (7af46f462d40…), FinalLightCurve_WASP-12 b_2017-01-23.png (c2ae05d60e26…), FinalLightCurve_WASP-12 b_2017-01-23.csv (3ba9259441e9…)

Compute resources

Wall time 385.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)